

Sprog og oversættelse

Lexical analysis: Regular expressions and NFAs

Part 6: A horror story

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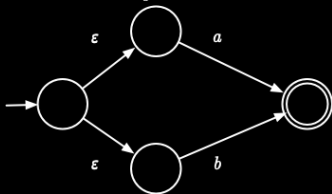
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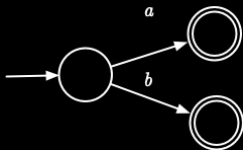


The ad hoc monster

Why do we have to learn all those difficult constructions? Why do we have to build a NFA for $a|b$ that is



?? Surely we could just draw



The ad hoc monster

- The constructions we meet in this course are the general techniques that are used in actual language implementation; they must be part of our toolbox.
- The ad hoc approach usually gets it *wrong*. The idea of leaving out epsilon transitions that one thinks one will not need is dangerous – there are relatively simple examples that show why this does not work.
- The constructions we meet in this course can always be used and always give us the correct results.

The confusion monster

Why are there so many different notations for regular expressions? It is so confusing! Can we use the SPO notation in the SS course? Can we use the SS notation in the SPO course?

In a course about compiler construction I would like to use a notation that is keyboard-friendly.

In a course about formal language it is best to use a notation that is closer to that of ordinary mathematics.

Learn to use the different notations and to keep them separated – they are different tools in our toolbox.